THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 12: **Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean mark of 10 students is 40 and of another 15 students is 60. The mean of all 25 students is: 1

(a) 50 **(b)** 1300 **(c)** 54 **(d)** 52

2. The mean of 59, 36, 46, 44, 50, 11 is 41. If 46 is replaced by 52, the new mean is: 1

(a) 42 **(b)** 41 **(c)** 49.67 **(d)** 47

3. The median of 46, 96, 29, 15, 31, 39, 70, 52 is: 1

(a) 47.25 **(b)** 40 **(c)** 42.5 **(d)** 31

4. The mean of 5 numbers is 28. The number to be added so that the mean becomes 30 is: 1

(a) 45 **(b)** 2 **(c)** 40 **(d)** 30

5. The mean of the first 9 multiples of 4 (that is 4, 8, ..., 36) is: 1

(a) 22 **(b)** 18 **(c)** 20 **(d)** 40

6. On a line graph, a town's population (in thousands) rises from 20 to 30. The percentage increase is: 1

(a) 50% **(b)** 60% **(c)** 10% **(d)** 25%

7. The mean mark of 20 students is 35 and of another 30 students is 45. The mean of all 50 students is: 1

(a) 40 **(b)** 41 **(c)** 43 **(d)** 2050

8. The mean of 20, 39, 32, 34, 52, 27 is 34. If 32 is replaced by 44, the new mean is: 1

(a) 46 **(b)** 36 **(c)** 41.33 **(d)** 34

9. The median of 49, 15, 16, 12, 98, 66, 92, 32 is: 1

(a) 40.5 **(b)** 98 **(c)** 47.5 **(d)** 33

10. The mean of 5 numbers is 30. The number to be added so that the mean becomes 25 is: 1

(a) 25 **(b)** 5 **(c)** -5 **(d)** 0

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The mean marks of 15 boys is 60 and of 10 girls is 70. Find the mean mark of the whole class of 25. 2

12. The mean of 5 observations 27, 38, 9, 33, 28 is 27. Find the mean if 6 is added to every observation. 2

13. Find the mean of the first 10 natural numbers that are multiples of 3. 2

14. The mean marks of 20 boys is 45 and of 30 girls is 55. Find the mean mark of the whole class of 50. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The mean weight of 10 students is 42 kg and of another 15 students is 48 kg. Find the mean weight of all of them. 3

16. The mean of 5 numbers is 26. If one number 49 is replaced by 59, find the new mean and the change in the mean. 3

17. The median of 60, 47, 76, 68, 36, 49, 21 is 49. If the largest value 76 is changed to 228, what happens to the median and the mean? 3

OR

The mean weight of 20 students is 36 kg and of another 30 students is 41 kg. Find the mean weight of all of them. 3

18. The mean of 5 numbers is 35. If one number 36 is replaced by 46, find the new mean and the change in the mean. 3

OR

The median of 41, 11, 49, 60, 43, 77, 33 is 43. If the largest value 77 is changed to 231, what happens to the median and the mean? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The temperatures (°C) at 8 am, 10 am, 12 noon, 2 pm, 4 pm and 6 pm were 18, 22, 27, 30, 26, 21. 5

(i) Find the rise or fall between each consecutive pair.

(ii) Find the mean temperature.

(iii) In which interval was the change largest?

OR

A class of 20 students has a mean mark of 50. Another 30 students with mean 60 join it. Find the mean of the combined group and the change in the original mean. 5

20. The temperatures (°C) at 8 am, 10 am, 12 noon, 2 pm, 4 pm and 6 pm were 20, 25, 31, 35, 30, 24. 5

(i) Find the rise or fall between each consecutive pair.

(ii) Find the mean temperature.

(iii) In which interval was the change largest?

OR

A class of 15 students has a mean mark of 64. Another 10 students with mean 74 join it. Find the mean of the combined group and the change in the original mean. 5

END OF QUESTION PAPER — **Total: 40 Marks**